

$\mathcal{K}_{0^+}^{\#1}$
 $\mathcal{K}_{0^-}^{\#1}$
 $\mathcal{K}_{0^+}^{\#1} \dagger$
 $\mathcal{K}_{0^-}^{\#1} \dagger^{\alpha\beta\chi}$

$\mathcal{K}_{0^+}^{\#1}$
 $\mathcal{K}_{0^-}^{\#1}$
 $\mathcal{K}_{0^+}^{\#1} \dagger$
 $\mathcal{K}_{0^-}^{\#1} \dagger^{\alpha\beta\chi}$

$-\frac{\Upsilon_1 k^2}{2}$	0				
0	$3k^2 \overset{(4)}{\kappa}_{15} + 3 \overset{(2)}{\kappa}_1$	$\mathcal{K}_{1^+}^{\#1}$	$\mathcal{K}_{1^+}^{\#2}$	$\mathcal{K}_{1^-}^{\#1}$	$\mathcal{K}_{1^-}^{\#2}$
		$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{1}{2} (\Upsilon_2 k^2 + 4 \overset{(2)}{\kappa}_1)$	$\frac{\Upsilon_3 k^2 + 4 \overset{(2)}{\kappa}_1}{2 \sqrt{2}}$	0
		$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$\frac{\Upsilon_3 k^2 + 4 \overset{(2)}{\kappa}_1}{2 \sqrt{2}}$	$\frac{1}{2} (\Upsilon_4 k^2 + 2 \overset{(2)}{\kappa}_1)$	0
		$\mathcal{K}_{1^-}^{\#1} \dagger^\alpha$	0	0	$-k^2 \overset{(4)}{\kappa}_2$
		$\mathcal{K}_{1^-}^{\#2} \dagger^\alpha$	0	0	$-\sqrt{2} k^2 \overset{(4)}{\kappa}_2$
					$\mathcal{K}_{2^+}^{\#1}$ $\mathcal{K}_{2^-}^{\#1}$
				$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$-\frac{\Upsilon_5 k^2}{2}$
				$\mathcal{K}_{2^-}^{\#1} \dagger^{\alpha\beta\chi}$	0

$\mathcal{J}_{0^+}^{\#1}$
 $\mathcal{J}_{0^-}^{\#1}$
 $\mathcal{J}_{0^+}^{\#1} \dagger$
 $\mathcal{J}_{0^-}^{\#1} \dagger^{\alpha\beta\chi}$

$\mathcal{J}_{0^+}^{\#1}$
 $\mathcal{J}_{0^-}^{\#1}$
 $\mathcal{J}_{0^+}^{\#1} \dagger$
 $\mathcal{J}_{0^-}^{\#1} \dagger^{\alpha\beta\chi}$

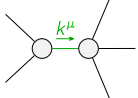
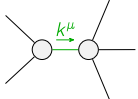
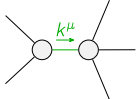
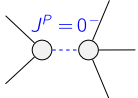
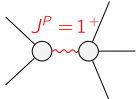
$\frac{1}{\text{Det}(0^+)}$	0				
0	$\frac{1}{3(k^2 \overset{(4)}{\kappa}_{15} + \overset{(2)}{\kappa}_1)}$	$\mathcal{J}_{1^+}^{\#1}$	$\mathcal{J}_{1^+}^{\#2}$	$\mathcal{J}_{1^-}^{\#1}$	$\mathcal{J}_{1^-}^{\#2}$
		$\mathcal{J}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{\Upsilon_4 k^2 + 2 \overset{(2)}{\kappa}_1}{2 \text{Det}(1^+)}$	$\frac{-\Upsilon_3 k^2 - 4 \overset{(2)}{\kappa}_1}{2 \sqrt{2} \text{Det}(1^+)}$	0
		$\mathcal{J}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$\frac{-\Upsilon_3 k^2 - 4 \overset{(2)}{\kappa}_1}{2 \sqrt{2} \text{Det}(1^+)}$	$\frac{\Upsilon_2 k^2 + 4 \overset{(2)}{\kappa}_1}{2 \text{Det}(1^+)}$	0
		$\mathcal{J}_{1^-}^{\#1} \dagger^\alpha$	0	0	$-\frac{1}{9k^2 \overset{(4)}{\kappa}_2}$
		$\mathcal{J}_{1^-}^{\#2} \dagger^\alpha$	0	0	$-\frac{\sqrt{2}}{9k^2 \overset{(4)}{\kappa}_2}$
					$\mathcal{J}_{2^+}^{\#1}$ $\mathcal{J}_{2^-}^{\#1}$
				$\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{1}{\text{Det}(2^+)}$
				$\mathcal{J}_{2^-}^{\#1} \dagger^{\alpha\beta\chi}$	0

Abbreviations used in matrices

$\Upsilon_1 == 3 \overset{(4)}{\kappa}_1 + 3 \overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6$
 $\&\& \Upsilon_2 == 4 \overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6$
 $\&\&$
 $\Upsilon_3 == 4 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_4$
 $\&\& \Upsilon_4 == 2 \overset{(4)}{\kappa}_{15} - 9 \overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4$
 $\&\& \Upsilon_5 == \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6$
 $\&\&$
 $\text{Det}(0^+) == -\frac{1}{2} k^2 (3 \overset{(4)}{\kappa}_1 + 3 \overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6)$
 $\&\& \text{Det}(1^+) ==$
 $\frac{1}{8} (k^4 (18 \overset{(4)}{\kappa}_2 \overset{(4)}{\kappa}_3 - 2 \overset{(4)}{\kappa}_3^2 - 2 \overset{(4)}{\kappa}_3 \overset{(4)}{\kappa}_4 - \overset{(4)}{\kappa}_4^2 + 2 (-9 \overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4) \overset{(4)}{\kappa}_6 + 4 \overset{(4)}{\kappa}_{15} (-18 \overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6)) +$
 $4 k^2 (-18 \overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6) \overset{(2)}{\kappa}_1$
 $\&\& \text{Det}(2^+) == -\frac{1}{2} k^2 (\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6)$

Lagrangian

$\overset{(2)}{\kappa}_1 \mathcal{K}_{\alpha\beta\chi} \mathcal{K}^{\alpha\beta\chi} - 2 \overset{(2)}{\kappa}_1 \mathcal{K}^{\alpha\beta\chi} \mathcal{K}_{\beta\alpha\chi} + \overset{(4)}{\kappa}_1 \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta + \overset{(4)}{\kappa}_2 \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta + \overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\alpha\chi} +$
 $\overset{(4)}{\kappa}_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\beta\chi} + \overset{(4)}{\kappa}_6 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\chi\alpha} - 6 \overset{(4)}{\kappa}_2 \partial^\chi \mathcal{K}^\alpha_{\alpha}{}^\beta \partial_\delta \mathcal{K}^{\delta}_{\chi\beta} - \frac{9}{2} \overset{(4)}{\kappa}_2 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\beta\chi} +$
 $\frac{1}{2} \overset{(4)}{\kappa}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\beta\chi} + \frac{1}{2} \overset{(4)}{\kappa}_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}_{\beta\chi} + \overset{(4)}{\kappa}_{15} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} - 2 \overset{(4)}{\kappa}_{15} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi}$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{1^+}^{\#1\alpha} == \mathcal{J}_{1^+}^{\#2\alpha}$	3	$\partial_\chi \partial^\alpha \mathcal{J}^\beta_{\beta}{}^\chi + \partial_\chi \partial^\chi \mathcal{J}^\beta_{\beta}{}^\alpha == 0$	
$\mathcal{J}_{2^-}^{\#1\alpha\beta\chi} == 0$	5	$3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta}{}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} +$ $4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^{\delta}{}_\epsilon{}^\epsilon + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\alpha}{}_\delta ==$ $3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha}{}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} +$ $2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta}{}_\epsilon{}^\epsilon + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\beta}{}_\delta$	
Total # constraint(s):	8		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$-\frac{63 \overset{(4)}{\kappa}_2^2 + 12 \overset{(4)}{\kappa}_2 \overset{(4)}{\kappa}_3 \overset{(4)}{\kappa}_6 - (\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6)^2}{\overset{(4)}{\kappa}_2 (\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6) (-18 \overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6)}$
	1	0	$-\frac{108 \overset{(4)}{\kappa}_2^2 - 6 \overset{(4)}{\kappa}_2 (\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6) - \sqrt{(-18 \overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6)^2 (92 \overset{(4)}{\kappa}_2^2 - 8 \overset{(4)}{\kappa}_2 (\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6) + (\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6)^2)}}{\overset{(4)}{\kappa}_2 (\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6) (-18 \overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6)}$
	1	0	$-\frac{108 \overset{(4)}{\kappa}_2^2 - 6 \overset{(4)}{\kappa}_2 (\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6) + \sqrt{(-18 \overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6)^2 (92 \overset{(4)}{\kappa}_2^2 - 8 \overset{(4)}{\kappa}_2 (\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6) + (\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6)^2)}}{\overset{(4)}{\kappa}_2 (\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6) (-18 \overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6)}$
	1	$-\frac{\overset{(2)}{\kappa}_1}{\overset{(4)}{\kappa}_{15}}$	$-\frac{1}{3 \overset{(4)}{\kappa}_{15}}$
	3	$\frac{4 (-18 \overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6) \overset{(2)}{\kappa}_1}{72 \overset{(4)}{\kappa}_{15} \overset{(4)}{\kappa}_2 + 2 \overset{(4)}{\kappa}_3^2 + 2 \overset{(4)}{\kappa}_3 \overset{(4)}{\kappa}_4 + \overset{(4)}{\kappa}_4^2 - 2 (\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4) \overset{(4)}{\kappa}_6 + 18 \overset{(4)}{\kappa}_2 (-\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6) - 4 \overset{(4)}{\kappa}_{15} (\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6)}$	$\frac{2 (324 \overset{(4)}{\kappa}_2^2 + 6 \overset{(4)}{\kappa}_3^2 + 8 \overset{(4)}{\kappa}_3 \overset{(4)}{\kappa}_4 + 3 \overset{(4)}{\kappa}_4^2 - 36 \overset{(4)}{\kappa}_2 (2 \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4) - 4 \overset{(4)}{\kappa}_3 \overset{(4)}{\kappa}_6 - 4 \overset{(4)}{\kappa}_4 \overset{(4)}{\kappa}_6 + 2 \overset{(4)}{\kappa}_6^2)}{(-18 \overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6) (-2 \overset{(4)}{\kappa}_3^2 - 2 \overset{(4)}{\kappa}_3 \overset{(4)}{\kappa}_4 - \overset{(4)}{\kappa}_4^2 + 18 \overset{(4)}{\kappa}_2 (\overset{(4)}{\kappa}_3 - \overset{(4)}{\kappa}_6) + 2 \overset{(4)}{\kappa}_3 \overset{(4)}{\kappa}_6 + 2 \overset{(4)}{\kappa}_4 \overset{(4)}{\kappa}_6 + 4 \overset{(4)}{\kappa}_{15} (-18 \overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_6))}$
Resolved unitarity condition(s):	(Demonstrably impossible)		